40th IEEE/ACM International Conference on Automated Software Engineering

IMPROVING NLSAT FOR NONLINEAR REAL ARITHMETIC

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Code available: `https://github.com/yogurt-shadow/clauseSMT`

Introduction

The Model-Constructing Satisfiability Calculus (MCSAT) framework has been applied to SMT problems over various arithmetic theories. NLSAT, an implementation using cylindrical algebraic decomposition (CAD) for explanation, is especially competitive for nonlinear real arithmetic (NRA) constraints. However, current Conflict-Driven Clause Learning (CDCL)-style algorithms only consider literal information when making decisions, and thus ignore the influence of clauses on arithmetic variables. This limitation may lead NLSAT to encounter unnecessary conflicts due to suboptimal literal choices. To address this issue, we analyze conflicts caused by literal decisions and incorporate clause-level information that directly affects arithmetic variables. We propose two main algorithmic improvements: a clause-level feasible-set-based look-ahead mechanism and an arithmetic propagation-based branching heuristic. We implement our solver, named clauseSMT, based on a dynamic variable ordering framework. Experiments indicate that clauseSMT is competitive on nonlinear real arithmetic problems compared with existing SMT solvers (CVC5, Z3, YICES2), and it outperforms all of them on satisfiable instances of SMT(QF_NRA) in SMT-LIB. We also evaluate the effectiveness of our proposed methods.

Look-Ahead Mechanism

Example 5.

$$\left.\begin{array}{l} c_1 : [-4,-2] \vee \boxed{[2,4]} \rightarrow \{[-4,-2] \cup [2,4]\}, \\ c_2 : [-6,-5] \vee \boxed{[1,5]} \rightarrow \{[-6,-5] \cup [1,5]\} \end{array}\right\} \bigwedge \rightarrow \{[2,4]\}$$

This example shows a path case, since the clauses can be satisfied by deciding literals in the green boxes.

Example 6.

$$\left.\begin{array}{l} c_1 : \boxed{[-4,-2]} \vee \boxed{[2,4]} \rightarrow \{[-4,-2] \cup [2,4]\}, \\ c_2 : \boxed{[-6,-5]} \vee \boxed{[5,6]} \rightarrow \{[-6,-5] \cup [5,6]\} \end{array}\right\} \bigwedge \rightarrow \varnothing$$

This example shows a block case: the clauses cannot be satisfied regardless of which literals we decide.

Figura 1: Literal Decision Cases

In our approach, we employ a look-ahead mechanism that first selects a pre-appointed value from the feasible-set. This pre-appointed value is then used to guide the search for a consistent decision path. This ensures that the feasible-sets of all decided literals during the processing procedure intersect at the clause-set level, allowing the arithmetic variable to be assigned the pre-appointed value. In the block case, the processing algorithm behaves identically to NLSAT, eventually triggering the resolve procedure to revise previous arithmetic assignments.

Clause-Level Propagation

Definition

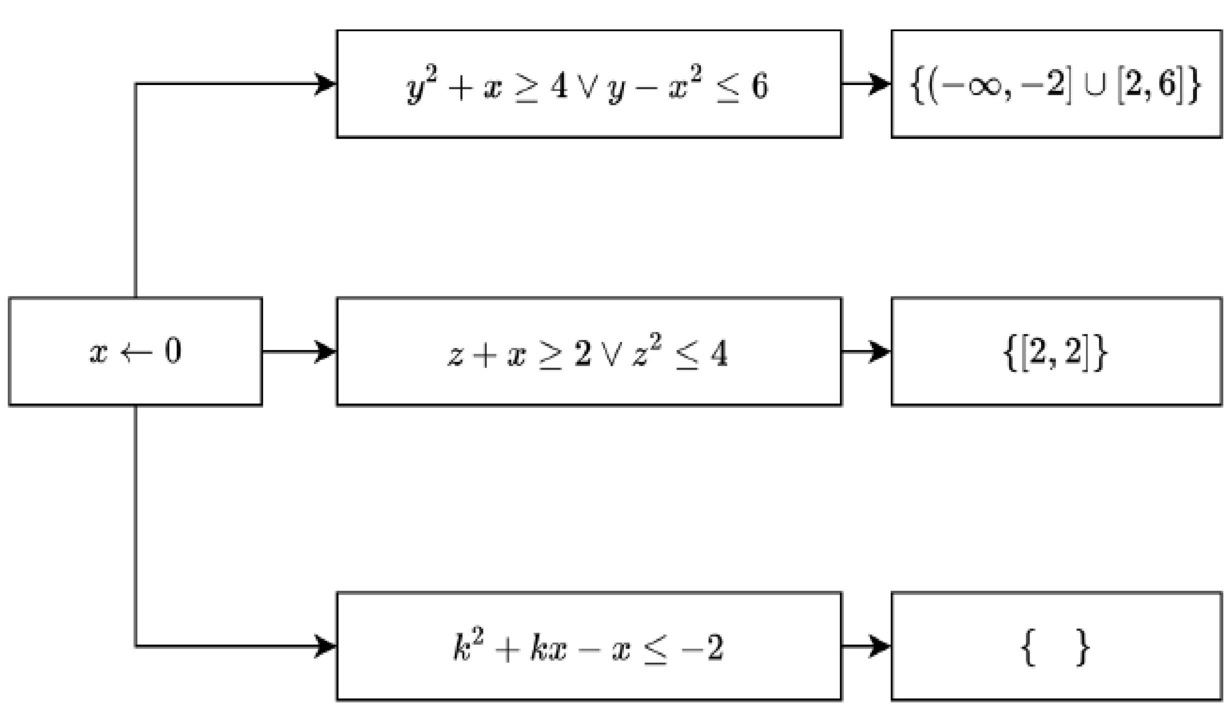

Figura 2: Clause-Level Propagation Cases

Branching Heuristic

We record information about fixed and blocked variables and prioritize them in the branching heuristic, ensuring that the search addresses these critical variables as early as possible

Algorithm 6: Propagation-Based Branching Heuristic

Output: A variable v to branch on

`// Prioritize blocked variables (potential conflicts) first`

if $blocked\_vars \neq \emptyset$ **then**

  $v \leftarrow$ select_from($blocked\_vars$);

`// Next, consider fixed variables (propagate value)`

else if $fixed\_vars \neq \emptyset$ **then**

  $v \leftarrow$ select_from($fixed\_vars$);

`// Otherwise, use VSIDS heuristic for normal variables`

else

  $v \leftarrow$ vsids_select();

return v

Figura 3: Branching Heuristic

Evaluation

The standard benchmark for evaluating SMT solvers is SMT-LIB.

Category	#inst		Z3	YICES2	CVC5	NLSAT	Ours
20161105-Sturm-MBO	405	SAT	0	0	0	0	0
		UNSAT	124	**285**	**285**	44	39
		SOLVED	124	**285**	**285**	44	39
20161105-Sturm-MGC	9	SAT	**2**	0	0	**2**	**2**
		UNSAT	**7**	0	0	**7**	6
		SOLVED	**9**	0	0	**9**	8
20170501-Heizmann	69	SAT	**2**	0	1	1	**2**
		UNSAT	1	12	9	10	**19**
		SOLVED	3	12	10	11	**21**
20180501-Economics-Mulligan	135	SAT	**93**	91	89	**93**	92
		UNSAT	39	39	35	**41**	**41**
		SOLVED	132	130	124	**134**	131
2019-ezsmt	63	SAT	56	52	50	**58**	36
		UNSAT	**2**	**2**	**2**	**2**	**2**
		SOLVED	58	54	52	**60**	38
20200911-Pine	245	SAT	234	**235**	199	**235**	234
		UNSAT	6	**8**	5	7	5
		SOLVED	240	**243**	204	242	239
20211101-Geogebra	112	SAT	**110**	99	91	**110**	98
		UNSAT	0	0	0	0	0
		SOLVED	**110**	99	91	**110**	98
20220314-Uncu	225	SAT	69	**70**	62	68	**70**
		UNSAT	**155**	153	148	**155**	152
		SOLVED	**224**	223	210	223	222
hong	20	SAT	0	0	0	0	0
		UNSAT	8	**20**	**20**	12	14
		SOLVED	8	**20**	**20**	12	14
hycomp	2752	SAT	**307**	227	225	244	291
		UNSAT	**2242**	2201	2212	2088	2181
		SOLVED	**2549**	2428	2437	2332	2472
kissing	45	SAT	**33**	10	17	12	14
		UNSAT	0	0	0	0	0
		SOLVED	**33**	10	17	12	14
LassoRanker	821	SAT	167	122	**305**	220	302
		UNSAT	151	260	**470**	174	311
		SOLVED	318	382	**775**	394	613
meti-tarski	7006	SAT	**4391**	4369	4343	**4391**	4372
		UNSAT	2605	2588	2581	**2611**	2588
		SOLVED	6996	6957	6924	**7002**	6960
UltimateAutomizer	61	SAT	35	39	35	**45**	39
		UNSAT	11	12	10	**13**	12
		SOLVED	46	51	45	**58**	51
zankl	166	SAT	**70**	58	58	62	56
		UNSAT	28	**32**	**32**	27	30
		SOLVED	**98**	90	90	89	86
Total	12134	SAT	5569	5372	5475	5541	**5608**
		UNSAT	5379	5612	**5809**	5191	5397
		SOLVED	10948	10984	**11284**	10732	11005

Figura 4: Experiment Result

About Author

The author has been accepted to the PhD program at the University of Toronto but is unable to enroll on time due to security clearance delays. If you are aware of any PhD opportunities or potential collaborations, he would be happy to hear from you.

Paper PDF

Video

Author's homepage

I strongly recommend taking a photo.